# Bayesian Optimisation

Proposed Report Structure and Group Responsibilities

Group 14

**Purpose.** This is a planning scaffold for the group to review and adapt. Each member will take primary responsibility for one section, develop expertise in that area, and lead the corresponding part of the presentation. Everyone should understand the overall argument and review another member's section.

**Central question.** How should we choose the next experiment to find a function's maximum when evaluations are expensive?

**Scope.** We study continuous, noise-free, black-box functions on a bounded one-dimensional interval, using a fixed evaluation budget. We use a Gaussian process (GP) with a squared-exponential kernel and fixed hyperparameters, derive expected improvement (EI), and compare it with an upper-confidence-bound (UCB) rule and uniform random search.

**Length and timing.** The suggested allocations below give approximately 20–22 pages of main text and a 15-minute presentation. The complete report must remain within the 30-page maximum. Presentation timings are provisional and should be refined during rehearsal.

## 1 Introduction and Motivation: Finding a Maximum When Experiments Are Expensive

**Section lead: Rhys** **Reviewer: Tom**

**Suggested report length:** 3 pages.
**Presentation allocation:** 2 minutes.

**Purpose:** Establish why the problem matters, explain the project's aims, and give readers the intuition needed for the mathematical sections.

- Introduce a concrete application where evaluating a function costs time or money, supported by scholarly sources.
- Explain the central problem: choosing inputs sequentially under a limited evaluation budget.
- Distinguish finding a maximum from accurately learning the entire function.
- Introduce exploration versus exploitation.
- Outline the Bayesian optimisation loop: update the model, choose the next input, evaluate the objective, and repeat.
- Define the assumptions and boundaries of our investigation.
- State the project aims and provide a roadmap of the report.

**Presentation responsibility:** Open with the motivating problem and explain what Bayesian optimisation is trying to achieve.

**Suggested output:** A motivating example and a diagram of the Bayesian optimisation loop.

## 2 Gaussian Processes: Modelling an Unknown Function and Its Uncertainty

**Section lead: Tom** **Reviewer: Tony**

**Suggested report length:** 4 pages.
**Presentation allocation:** 2 minutes.

**Purpose:** Explain the probabilistic model that supplies predictions and uncertainty estimates to the acquisition rules.

- Introduce the GP prior, mean function, and squared-exponential kernel.
- State and interpret the posterior mean and variance equations obtained by conditioning on observations.
- Use a small worked example to connect the equations to a plot.
- Explain how the kernel length scale affects the model's assumptions about smoothness.
- Explain the meaning of a pointwise 95% posterior band and distinguish model-dependent uncertainty from guaranteed error bounds.
- Establish consistent notation for the observed data, posterior mean $\mu_n(x)$, and posterior standard deviation $\sigma_n(x)$.
- Keep the background focused on what is needed to understand Bayesian optimisation.

**Presentation responsibility:** Explain how to read the GP plot and how a new observation changes the model.

**Suggested output:** An annotated posterior plot and a worked conditioning example.

## 3 Choosing the Next Experiment: Expected Improvement and Upper Confidence Bounds

**Section lead: Tony** **Reviewer: Owen**

**Suggested report length:** 5–6 pages.
**Presentation allocation:** 4 minutes.

**Purpose:** Develop the main mathematical argument: how predictions and uncertainty are converted into a decision about the next evaluation.

- Define improvement relative to the best value observed so far.
- Derive expected improvement by integrating positive improvement against the Gaussian posterior.
- Interpret the resulting formula and explain its one-step decision-making interpretation.

- Analyse limiting behaviour, including the case of zero posterior uncertainty.
- Show how EI changes with posterior mean and uncertainty when the other quantity is held fixed.
- Introduce the UCB acquisition rule and explain the effect of its exploration parameter.
- Compare EI and UCB using candidate points evaluated on the same posterior.
- Explain how maximising an acquisition function selects the next input and completes one iteration of Bayesian optimisation.

**Presentation responsibility:** Explain the central EI derivation and interpret why EI and UCB may choose different points.

**Suggested output:** A clearly explained derivation and a worked comparison of acquisition decisions.

# 4 Educational Demonstration: Making Sequential Decisions Visible

**Section lead: Owen** **Reviewer: Ronald**

**Suggested report length:** 4 pages.
**Presentation allocation:** 5 minutes.

**Purpose:** Translate the mathematics into an interactive demonstration that helps the audience understand uncertainty and sequential decision-making.

- Define the intended audience and learning objectives: interpreting uncertainty, understanding exploration versus exploitation, and explaining next-point choices.
- Present a hidden function with multiple peaks, three initial observations, a GP posterior plot, an acquisition curve, and an evaluation budget.
- Let the audience choose the next input; reveal its function value and update the model.
- Display EI, mark its maximiser, evaluate the selected input, and repeat for a few steps.
- Switch to UCB on the same posterior and vary its exploration parameter to show how the proposed input changes.
- Reveal the hidden function and compare the best observation with the true maximum.
- Explain implementation choices that affect correctness, including dense-grid acquisition maximisation and which objective evaluations count towards the budget.
- Include annotated screenshots and explain how the design supports the learning objectives.
- Record rehearsal or audience feedback if collected, and distinguish observed feedback from intended learning benefits.
- Prepare a static sequence of figures as a fallback for the live demonstration.

**Presentation responsibility:** Lead the audience interaction and explain the audience's and algorithm's decisions.

**Suggested output:** A working demonstration, annotated screenshots, and a short presentation script with a static fallback.

## 5 Comparative Experiments, Limitations and Conclusions: When Does Bayesian Optimisation Help?

**Section lead: Ronald** **Reviewer: Rhys**

**Suggested report length:** 4–5 pages.
**Presentation allocation:** 2 minutes.

**Purpose:** Evaluate the methods fairly, interpret their behaviour, and answer the project's central question.

- Compare EI, UCB, and uniform random search on two contrasting test functions.
- Specify a reproducible protocol with equal evaluation budgets, shared initial observations, and repeated initial designs.
- Plot the best-observed value against evaluation count, including variability across runs.
- Investigate sensitivity to kernel length scale and the UCB exploration parameter.
- Interpret successes and failures using the earlier theory: unsuitable kernel assumptions, missed narrow peaks, and acquisition optimisation.
- Distinguish illustrative synthetic experiments from evidence about performance in costly real-world applications.
- Explain what a small evaluation budget cannot guarantee.
- Conclude by answering the opening question and identifying the main lessons and possible extensions.

**Presentation responsibility:** Present the main experimental finding, explain the limitations, and bring the presentation to a clear conclusion.

**Suggested output:** Comparative performance plots, a sensitivity analysis, and a concise answer to the central question.

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**Shared responsibilities and assessment requirements**

- **Scholarly evidence:** Each member should read and synthesise relevant scholarly sources. Agree a consistent referencing style and maintain a shared bibliography.
- **Mathematical understanding:** Everyone should understand the complete BO loop, the interpretation of the GP posterior, and the central EI argument.
- **Coherence:** Agree notation, terminology, figure styles, and the intended level of explanation before writing. Review the assembled report so that it reads in a single voice.
- **Connections between sections:** Coordinate the posterior examples used in Sections 2 and 3, the acquisition rules used in Sections 3 and 4, and the implementation and experimental protocol used in Sections 4 and 5.
- **Presentation preparation:** Each member leads their allocated part, rehearses the transitions, and prepares to answer questions about both their section and the overall project.
- **Supporting material:** Include a short unnumbered abstract, a bibliography, and only essential appendices. Submit supporting code and instructions alongside the report.

- **Individual reflections and AI use:** Include each member's signed reflection and AI-use statement at the end of the report, following the assessment guidance.